

Sub-Field Management-Zone Delineation Inside the Parcel

BUCKET 5 - FIELD / PARCEL BOUNDARY AND SUB-FIELD STRUCTURE

For EU arable farmers and agronomic co-ops applying variable-rate nitrogen and seed under CAP eco-schemes, we train a temporal segmentation model on multi-year Sentinel-1/2 time series inside known EuroCrops/LPIS parcel polygons to produce sub-field management zones that are optimised at training time for cross-season stability and predictive power on a held-out response variable, rather than single-date NDVI homogeneity.

VIABLE

All hard constraints pass, and the imagery/data/budget/pitch fundamentals are clean, but the H1 novelty margin is genuinely narrow and is being actively closed by an existing, currently-publishing research group (Calgary GIV) whose May 2026 review names this exact gap. Recommend proceeding only with the week-1/2 kill-criteria checks run first, and with the student prepared to sharpen or pivot the delta if a closing preprint appears.

Confidence: Medium. Would rise to High after (a) directly reading the discussion/future-work section of the Heidari, Khanmohammadi & Samavati 2026 systematic review in full to check for an already-announced follow-up study, and (b) empirically confirming a specific EU region with sufficient multi-year cloud-free Sentinel-2 and LPIS/EuroCrops coverage together.

Hard constraint check

Constraint	Verdict	Reasoning
H1 • Novelty	PASS	(narrow) — training-time stability optimisation plus a predictive-power probe is not published as such, but the margin over Jovanovic 2025 and Heidari 2024's post-hoc metrics is thin and actively being closed.
H2 • Not generic	PASS	not a banned pattern; the specific technical combination is not generic, though it sits in an active niche.
H3 • Imagery is primary input	PASS	Sentinel-1/2 satellite imagery is the primary and only model input; sensor, resolution, revisit and licence stated.
H4 • Geography & user	PASS	EU arable farmers and precision-ag agronomists/co-ops are named as the specific user segment.
H5 • Compute & budget	PASS	dataset size (~20-40GB), compute (a few hours on MPS, fits in 24GB), and cash cost (~RM0) are all comfortably within budget.
H6 • Pitch quality	PASS	the before/after stability visualisation is a clear, demoable 60-second story for a CS academic panel.

The problem

Once a parcel's outer boundary is known (now largely solved at global scale by Delineate Anything v2, arXiv:2607.19069), farmers and agronomists still need to know how to split that parcel into contiguous management zones for variable-rate input application. Current operational practice is single-date k-means/fuzzy-c-means clustering of NDVI or apparent soil electrical conductivity (e.g. Georgi et al. 2017, Martinez-Casasnovas et al. 2018), which produces zones that are unstable from year to year because they

are fit to one season's weather and phenology rather than to a persistent yield-limiting structure. This instability undermines the practical value of management zones: a zone map that changes every year cannot anchor a multi-year variable-rate application plan. Existing multi-year studies (Garg et al. 2022 SAMZ-Desert; Jovanovic et al. 2025) measure this instability post hoc with agreement statistics (Cohen's Kappa, Jaccard, Dice) but do not optimise the zoning method itself for stability, and do not tie zone quality to an explicit predictive criterion.

Proposed technical approach

Input: per-parcel multi-year stacks of Sentinel-1 (VV/VH, 10m) and Sentinel-2 (10 bands, 10-20m) pixel time series, clipped to individual EuroCrops/LPIS parcel polygons across 3-5 consecutive growing seasons. Architecture: a lightweight temporal encoder per pixel (a compact adaptation of Pixel-Set Encoder + Temporal Attention Encoder / U-TAE, Garnot & Landrieu 2021, arXiv:2107.07933) producing an embedding per pixel per season, followed by a differentiable clustering head (deep embedded clustering or soft k-means) assigning each pixel to one of K zones (K chosen per parcel via silhouette/elbow on a validation split). Objective: a composite loss with three terms — (a) within-season clustering compactness (standard intra-cluster variance term, as in classical MZ delineation); (b) a stability regularizer that computes zone-assignment agreement (a differentiable relaxation of Dice/Jaccard, or an entropy penalty on assignment drift) between the same parcel's zone maps in consecutive seasons after Hungarian-algorithm label matching; (c) a predictive term that regresses zone-mean multi-sensor features from season t against an independent proxy response variable observed in season t+1 (e.g. peak-season biomass/NDVI-integral as a proxy for yield, since real yield-monitor data is not available at this scale), scored via a linear probe. Training: self-supervised, no manual zone labels required; the response-variable proxy substitutes for real yield data and this substitution is an explicit, stated limitation, not a claim of yield prediction. Output: a stable, multi-year sub-field zone raster per parcel plus a per-parcel stability/predictive score.

Novelty

WHAT THE DELTA ACTUALLY IS

The defensible delta is narrow but real: existing management-zone literature (Georgi 2017, Martinez-Casasnovas 2018, Scudiero 2018, Garg 2022 SAMZ-Desert, Heidari & Samavati 2024, Jovanovic et al. 2025) all treat multi-year stability as something to measure after clustering, using post-hoc agreement statistics or anomaly-year detection — none of them make stability (and predictive power over a held-out response) an explicit differentiable training objective that shapes the segmentation itself. This candidate proposes that shift: a trained stability regularizer plus a predictive-power probe, with an ablation that isolates the regularizer's contribution from the representation. This is not 'applied X to Y country' because the technical core is a new training objective and evaluation protocol, not a re-run of an existing pipeline on new geography; the EU/LPIS setting is the data regime that makes multi-year parcel-level supervision-free training feasible, not incidental window dressing. However, the delta is genuinely narrow: the Calgary GIV group (Heidari, Khanmohammadi, Samavati) published a systematic review in May 2026 that explicitly identifies this exact gap ('shift validation efforts... toward multi-year yield stability assessments') as the field's next step, meaning the gap is real but is also being actively targeted by an existing, productive research group right now.

PRIOR ART THAT THREATENS IT

Heidari & Samavati 2024 (10.3390/agriculture14050688) already builds a dissimilarity metric to detect years that disrupt multi-year zone consistency — adjacent to, but not identical to, training-time stability optimisation. Heidari, Khanmohammadi & Samavati 2026 (10.3390/s26103249, Sensors, a systematic review of 137 papers 2000-2025) explicitly names multi-year stability assessment as the field's unmet validation need, confirming the gap is real as of May 2026 but also signalling this exact research group is positioned to close it next. Jovanovic et al. 2025 (10.3390/su17156931) already reports Kappa/Jaccard/

Dice-based 5-year stability assessment across three data sources, which is very close to this candidate's proposed evaluation protocol (though still post-hoc, not a training objective). Garg et al. 2022 SAMZ-Desert and Fontanet et al. 2020 'Dynamic Management Zones for Irrigation Scheduling' both already study temporal instability of zones empirically. None of these use a deep SITS encoder (PSE-TAE/U-TAE-style) or make stability a differentiable loss term, which is where this candidate's remaining novelty lives — but that margin could close quickly given the pace of the Calgary group's publications.

GENERIC-PATTERN CHECK

The core is not a chatbot/LLM assistant, not a dashboard/farm-management portal, not leaf-photo disease classification, not a plain NDVI viewer, not tabular yield-from-weather-and-soil, not a crop-type-classification rerun, and not a mobile app calling someone else's model API — none of the eight H2-banned patterns apply. Could the one-sentence summary describe 50 other FYPs? No: the specific combination of (EU LPIS/EuroCrops parcels) + (multi-year Sentinel-1/2 SITS) + (an explicit differentiable stability-and-predictive-power training objective, evaluated with an ablation against post-hoc validation) is a narrow technical target, not a generic template. It does, however, sit inside an active, fast-moving research niche (see prior_art), so genericity risk here is really 'incremental extension of a specific existing line of work' rather than 'templated X-to-Y'.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-2 MSI	10 m	5	B2,B3,B4,B5,B6,B7,B8,B8A,B11,B12 (10 bands, 10-20m resampled to 10m)	Copernicus free and open data licence
Sentinel-1 GRD (C-band SAR)	10 m	6-12 (current S1A+S1C constellation; degraded from nominal 6-day since S1B loss) [uncertain: exact current revisit for target AOI not independently reverified]	VV, VH	Copernicus free and open data licence

Sensor fusion of SAR + optical, both free and open, encouraged per H3.

Yes. The model's only input is per-pixel Sentinel-1/2 time series clipped to parcel polygons; the polygons themselves (from EuroCrops/LPIS) are a static mask, not the learning signal, and are treated as given per the candidate description (parcel delineation is the already-solved problem this candidate steps past). No non-imagery tabular data drives the model.

Datasets

EUROCROPS V2.0

Access route	DOI landing page https://doi.org/10.2905/b9fb9e67-78a9-4327-9d59-39a928d812d3 ; bulk download https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/DRLL/EuroCropsV2/
Auth required	No
Licence	CC BY 4.0
Approx. size	Full dataset ~47M parcels/21Mha across 18 countries and up to 16 years; a single-country, multi-year subset needed for this project is a few GB of vector data
Accessible from MY	Yes (public JRC FTP, no geofencing found)
How verified	WebFetch of essd.copernicus.org/articles/18/4075/2026/ (EuroCrops v2.0 ESSD paper), confirming GeoParquet format, EPSG:3035, CC BY 4.0, and the JRC FTP download route

AI4BOUNDARIES

Access route	JRC Open Data Catalogue http://data.europa.eu/89h/0e79ce5d-e4c8-4721-8773-59a4acf2c9c9 (FTP access also listed); GitHub companion github.com/waldnerf/ai4boundaries
Auth required	No
Licence	Open (JRC open data); exact licence string not independently re-verified in this pass [uncertain]
Approx. size	7,831 Sentinel-2 samples of 256x256px at 10m plus 512x512px aerial orthophotos, 2.5M parcels across 47,105 sq km — order of a few GB for the Sentinel-2-only subset needed here
Accessible from MY	Yes (open FTP/catalogue, no login found)
How verified	WebSearch confirming ESSD paper (essd.copernicus.org/articles/15/317/2023/) and JRC catalogue URL; direct WebFetch of the catalogue page failed (dataset-not-found on the specific handle tried) so the exact current URL should be re-confirmed in week 1 [uncertain]

SENTINEL-1/2 IMAGERY

Access route	Copernicus Data Space Ecosystem (dataspace.copernicus.eu) via STAC / openEO / Sentinel Hub API, no payment required; optional Microsoft Planetary Computer STAC mirror
Auth required	Free registration for API/bulk access; browser access requires no account
Licence	Copernicus free and open data policy
Approx. size	Only parcel-clipped time series are needed (not full scenes); estimated 10-20GB for a multi-year, single-region experiment covering several thousand parcels
Accessible from MY	Yes
How verified	WebFetch of dataspace.copernicus.eu confirming free registration-optional access and STAC/openEO/Sentinel Hub API routes

TOTAL DATA VOLUME

Approximately 20-40GB for a minimum viable single-country, 3-5 year experiment (parcel-clipped Sentinel-1/2 time series + EuroCrops vector layer + AI4Boundaries validation subset). Well under the ~200GB H5 practical ceiling.

LABELS AND GROUND TRUTH

No official ground truth exists for 'correct' management zones anywhere in the literature — this is inherent to the task, not a gap specific to this project. Supervision substitutes are: (1) EuroCrops/LPIS parcel polygons as a hard spatial mask (real, official data); (2) a proxy response variable (independent-season biomass/NDVI-integral) used only to score predictive power, not as a training label for the zones themselves; (3) the classical k-means/NDVI baseline as a comparison point, not ground truth. Risk: because no external validity check exists beyond stability and proxy-correlation, evaluation is at some risk of being self-referential — a concern shared by all management-zone-delineation literature to date (see Heidari, Khanmohammadi & Samavati 2026 systematic review, which explicitly calls out the field's weak validation practices).

Baselines to beat

Paper	Reported metric	Dataset
Automatic delineation algorithm for site-specific management zones based on satellite remote sensing data Precision Agriculture (Springer) 2017 · 10.1007/s11119-017-9549-y	Automated single-date multispectral (RapidEye) MZ segmentation, qualitative validation against relative yield expectation	RapidEye imagery, own field trials
Workflow to Establish Time-Specific Zones in Precision Agriculture by Spatiotemporal Integration of Plant and Soil Sensing Data Agronomy (MDPI) 2018 · 10.3390/agronomy8110253	Fuzzy cluster analysis of soil ECa + NDVI, time-specific zone workflow, qualitative	21-ha maize field, Italy, 2 years
Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks (U-TAE / PSE-TAE, PASTIS dataset) ICCV 2021 · arXiv: 2107.07933 / DOI 10.1109/iccv48922.2021.00483	State-of-the-art panoptic/semantic segmentation of parcel-level SITS	PASTIS (France)

A New Dissimilarity Metric for Anomaly Detection in Management Zones Delineation Constructed from Time-Varying Satellite Images Agriculture (MDPI) 2024 · 10.3390/agriculture14050688	Zoning Dissimilarity Metric (ZDM) identifies anomalous years disrupting multi-year zone consistency	Sentinel-2, 39 fields, Canada
Multi-Source Data Integration for Sustainable Management Zone Delineation in Precision Agriculture Sustainability (MDPI) 2025 · 10.3390/su17156931	Cohen's Kappa, Jaccard and Dice agreement of MZs across a 5-year period (2018-2022), Kappa > 0.90 for best data source	EM38 soil conductivity + soil chemistry + Sentinel-2 indices, own field

Evaluation protocol

Primary metrics: (1) cross-season zone stability, measured as Cohen's Kappa / Jaccard / Dice between each parcel's zone map in consecutive seasons after Hungarian-algorithm label re-matching (directly comparable to Jovanovic et al. 2025's post-hoc metric, but here computed as, and optimised as, part of the training objective rather than only evaluated after the fact); (2) predictive power, measured as R^2 / correlation between season-t zone-mean spectral-SAR features and an independent proxy response variable in season t+1, using spatial block cross-validation (parcels held out by geographic block, not randomly, to avoid leakage). Success criterion: statistically significant improvement in both stability (higher Kappa) and predictive R^2 over (a) a single-date k-means/NDVI baseline reproducing Georgi et al. 2017's approach and (b) an ablation of the proposed model with the stability regularizer removed (isolating the regularizer's actual contribution from generic representation quality). Test split: held-out parcels from a geographic sub-region not seen in training, within the same country, to test spatial generalisation; held-out final season to test temporal generalisation.

Feasibility

COMPUTE

Peak memory	Estimated 4-10GB for per-parcel patch-based training (small spatial patches, e.g. up to 64x64 px, x T timesteps x modest batch size) with a PSE-TAE/U-TAE-scale encoder; comfortably within 24GB unified memory
Est. training hours (MPS)	Estimated 2-6 hours per full training run on a single-region, multi-year dataset of a few thousand parcels, given the model is much smaller than the segmentation-only PASTIS baselines it borrows from
Fits 24 GB	Yes
MPS-compatible ops	Standard conv1d/conv2d, multi-head attention, and batchnorm are MPS-supported in current PyTorch; risk: the original PSE-TAE/U-TAE reference implementations use some custom CUDA-only ops (e.g. certain sparse-set aggregation kernels) that would need reimplementaion in native PyTorch ops for MPS — this is a real but manageable engineering task, not a blocker

Colab/Kaggle fallback	No, but available as a backup if MPS reimplementations of any custom ops prove difficult
Notes	The dominant compute cost is data preparation (clipping Sentinel-1/2 time series to parcel polygons via STAC), not model training; this should be budgeted as the larger time risk in week 1-2.
Cash cost	Approximately RM 0. All imagery (Copernicus Data Space Ecosystem), parcel data (EuroCrops CC BY 4.0, AI4Boundaries JRC open data), and compute (own MacBook Pro M4 Pro; Colab free tier as optional backup) are free-tier or open licence. No commercial VHR imagery or paid API is required.
Build : research	Approximately 65:35 engineering:research. Engineering: the parcel-clipped Sentinel-1/2 data pipeline (STAC querying, cloud masking, compositing), the training pipeline for the temporal encoder and clustering head, and the interactive demo/visualisation app. Research: designing and validating the stability regularizer and predictive-power probe, running the ablations that isolate their contribution, and engaging honestly with the closely adjacent Calgary-group literature.
Publication venue fit	Computers and Electronics in Agriculture or Remote Sensing (MDPI) fit well given the topic sits squarely in precision-agriculture remote sensing with a methodological contribution; CVPR EarthVision or ICLR ML4RS are plausible if the SITS-encoder/differentiable-stability-loss angle is emphasised over the agronomy angle. IGARSS is a realistic fallback given the SAR+optical fusion component.

DEMO ARTIFACT

A laptop demo showing, for one or a handful of chosen EU parcels: (1) the parcel polygon on a basemap; (2) the classical single-date k-means/NDVI zoning for each of 3-5 seasons, visibly flickering/changing shape year to year; (3) the proposed model's zoning for the same seasons, visibly stable in shape; (4) a live-updating stability score (Kappa) comparing the two. Runs in under 2 minutes on cached, pre-downloaded parcel time series (no live satellite fetch needed during the demo itself).

User and pain

USER SEGMENT

EU arable farmers and the agronomists/co-ops serving them in CAP-eligible grain- and row-crop-growing regions with good multi-year LPIS/EuroCrops and cloud-free Sentinel-2 coverage (e.g. parts of France, Germany, Denmark, or the Netherlands — exact region to be confirmed in week 1 by checking cloud-free revisit and multi-year LPIS availability together). These are the actual buyers of variable-rate application equipment and precision-ag advisory services today. Not 'farmers globally'.

QUANTIFIED PAIN

Precision farming adoption today is concentrated among large farms in developed countries and remains limited elsewhere (Finger, Swinton, El Benni & Walter 2019, Annual Review of Resource Economics, DOI 10.1146/annurev-resource-100518-093929), meaning most EU arable land still uses uniform (non-zoned) input application despite the demonstrated agronomic case for zoning. **[uncertain]** A precise EU-wide euro or hectare figure for the annual cost of manual/soil-sampling-based zone delineation, or the specific fertiliser-saving percentage attributable to zoning versus uniform application, was not found in the sources reviewed in this pass and should be sourced directly (e.g. from a CAP eco-scheme monitoring report or an agronomy extension source) in week 1 before finalising the pitch's headline number.

Key risks

1. Novelty erosion: the Calgary GIV group (Heidari, Khanmohammadi, Samavati) has explicitly identified this exact gap in a May 2026 systematic review and is an active, publishing group in precisely this niche — real risk of being scooped or of the delta shrinking further before the FYP concludes.
2. Circular/weak evaluation: with no true management-zone ground truth existing anywhere in the field, both the stability metric and the predictive-power proxy are self-referential to some degree; a rigorous panel could challenge whether 'better' zones are actually agronomically meaningful.
3. MPS/engineering risk: reference SITS-encoder implementations (PSE-TAE/U-TAE) include some CUDA-specific kernels that will need reimplementation for Apple Silicon MPS; scope and time for this should be estimated early.
4. Parcel-size risk: many EU LPIS parcels are small enough (especially outside large-farm grain regions) that meaningful sub-field zoning is not possible at 10m resolution; the usable parcel pool may be smaller than expected, narrowing the study region.
5. Data-availability risk: sufficient cloud-free Sentinel-2 revisit across 3-5 consecutive growing seasons, aligned with multi-year LPIS/EuroCrops coverage, needs to be confirmed for a specific candidate region before committing — this is not guaranteed everywhere in the EuroCrops footprint.

Kill criteria

1. Week 1-2: (1) Pick a candidate EU region and confirm it has ≥ 3 consecutive years of both EuroCrops/LPIS coverage and sufficient cloud-free Sentinel-2 revisit for compositing; if not, pick another region or abandon.
2. Run the classical k-means/NDVI baseline on a sample of parcels in that region and measure its actual year-to-year Kappa; if the baseline is already highly stable (e.g. Kappa > 0.7), there is no headroom for the proposed method to improve on, and the project should pivot or stop.
3. Re-search arXiv/OpenAlex for a 2026 preprint from the Calgary GIV group or others that already implements stability as a training objective (not just a validation metric); if found, either differentiate further or abandon.

Open / unverified items

- Exact euro/hectare figure for the cost of manual management-zone delineation or the fertiliser-saving percentage attributable to zoning in the EU (H6 quantified_pain number needs primary sourcing in week 1).
- Whether the Calgary GIV group (Heidari, Khanmohammadi, Samavati) has an unpublished or in-press 2026 preprint that already implements stability as a training objective rather than a post-hoc metric.
- Exact current Sentinel-1 revisit cadence for a specific candidate EU AOI given the post-S1B constellation status.
- AI4Boundaries' exact current licence string and whether its originally-cited JRC catalogue URL is still live (direct WebFetch attempt failed; access route should be re-verified in week 1).
- Whether a specific EU region can be found with simultaneously sufficient multi-year cloud-free Sentinel-2 revisit and multi-year LPIS/EuroCrops coverage — not yet empirically confirmed for any single named region.
- Exact MPS-compatibility of all ops in the reference PSE-TAE/U-TAE implementations (some custom CUDA kernels may require reimplementation).

Key references

Automatic delineation algorithm for site-specific management zones based on satellite remote sensing data

Precision Agriculture · 2017 · [10.1007/s11119-017-9549-y](#)

Named baseline: canonical single-date k-means/NDVI automatic management-zone delineation method this candidate must beat.

A New Dissimilarity Metric for Anomaly Detection in Management Zones Delineation Constructed from Time-Varying Satellite Images

Agriculture (MDPI) · 2024 · [10.3390/agriculture14050688](#)

Closest prior art on multi-year zone consistency; defines the post-hoc evaluation style this candidate must clearly differentiate from.

The Evolution of Data-Driven Management Zone Delineation: A Systematic Review

Sensors (MDPI) · 2026 · [10.3390/s26103249](#)

May 2026 systematic review that explicitly names 'multi-year yield stability assessment' as the field's key unmet validation need — both confirms the gap and signals active competition.

Multi-Source Data Integration for Sustainable Management Zone Delineation in Precision Agriculture

Sustainability (MDPI) · 2025 · [10.3390/su17156931](#)

Already reports Kappa/Jaccard/Dice multi-year stability assessment across 5 years and 3 data sources — the evaluation protocol this candidate's metric must be shown to improve on, not merely replicate.

Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks

ICCV · 2021 · [arXiv:2107.07933](#)

Source architecture (PSE-TAE/U-TAE) and PASTIS dataset for the parcel-level SITS encoder this candidate adapts for management-zone embedding.

EuroCrops v2.0: multi-annual harmonized parcel level crop type data linked to European Union-wide survey, statistical, and Earth Observation products

Earth System Science Data · 2026 · [essd.copernicus.org/articles/18/4075/2026/](#)

Verified, current access route, size, and licence for the multi-year LPIS parcel polygons this candidate depends on.

Workflow to Establish Time-Specific Zones in Precision Agriculture by Spatiotemporal Integration of Plant and Soil Sensing Data

Agronomy (MDPI) · 2018 · [10.3390/agronomy8110253](#)

Early recognition that management zones should be time-specific rather than static — conceptual precursor this candidate operationalises with deep learning.

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